12

Partitions, File Systems, and Swap

Objectives

After completing this lesson, you should be able to:

- Describe disk partitioning
- Use disk partitioning utilities
- List supported file system types
- Explain file system creation, mounting, and maintenance
- Describe block volumes in Oracle Cloud Infrastructure
- Describe access control lists (ACLs)
- Describe and maintain swap space

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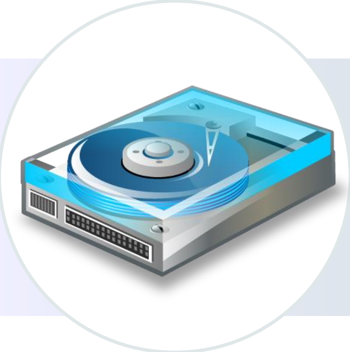
Disk Partitions

- Partitioning divides a disk drive into logical disks.
  - Each partition is treated as a separate disk.
  - A partition table defines the partitions.
- Minimum recommended partitions (file systems):
  - `/` (root)
  - `/boot`
  - `swap`
- Create additional partitions to simplify administration.
- Extended partitions allow the creation of more than four primary partitions.



Partition Layout: Example

- Oracle Linux virtual machine is installed and running under Oracle VM.
- Three primary partitions are created during installation:
  - `xvda1`, mounted on `/boot`
  - `xvda2`, mounted on `/`
  - `xvda3`, mounted on `/home`
- A fourth partition, `xvda4`, is created as an extended partition.
  - Extended partitions can be subdivided into logical partitions.
- One logical partition, `xvda5`, is created on the extended partition.
  - This logical partition is designated as a `swap` partition.
- Partition devices are listed in the `/proc/partitions` file:

```
# cat /proc/partitions
```

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Partition Table Manipulation Utilities

- Three partition utilities are presented in this lesson:
  - `fdisk`
  - `cfdisk`
  - `parted`
- Do not partition a device while it is in use.
- Ensure that file systems are unmounted.
  - Use the `umount` command.
- Ensure that swap space is disabled.
  - Use the `swapoff` command.

The `fdisk` Utility

- The `fdisk` utility is a partition table manipulator for Linux.
- Use the `fdisk -l` option to list the partition table.
  - **Device:** Lists the partitions
  - **Boot:** `*` indicates that the partition contains boot files
  - **Start and End:** The starting and ending sectors
  - **Blocks:** The number of blocks allocated to the partition
  - **Id and System:** The partition type
- Partition naming (example: `/dev/xvda1`)
  - `/dev`: The directory containing device files
  - `sd`: SCSI disk; `hd`: IDE disk; `xvd`: Virtual disk
  - `a`: First disk; `b`: Second disk; `c`: Third disk
  - `1`: First partition; `2`: Second partition; `3`: Third partition

Using the `fdisk` Utility

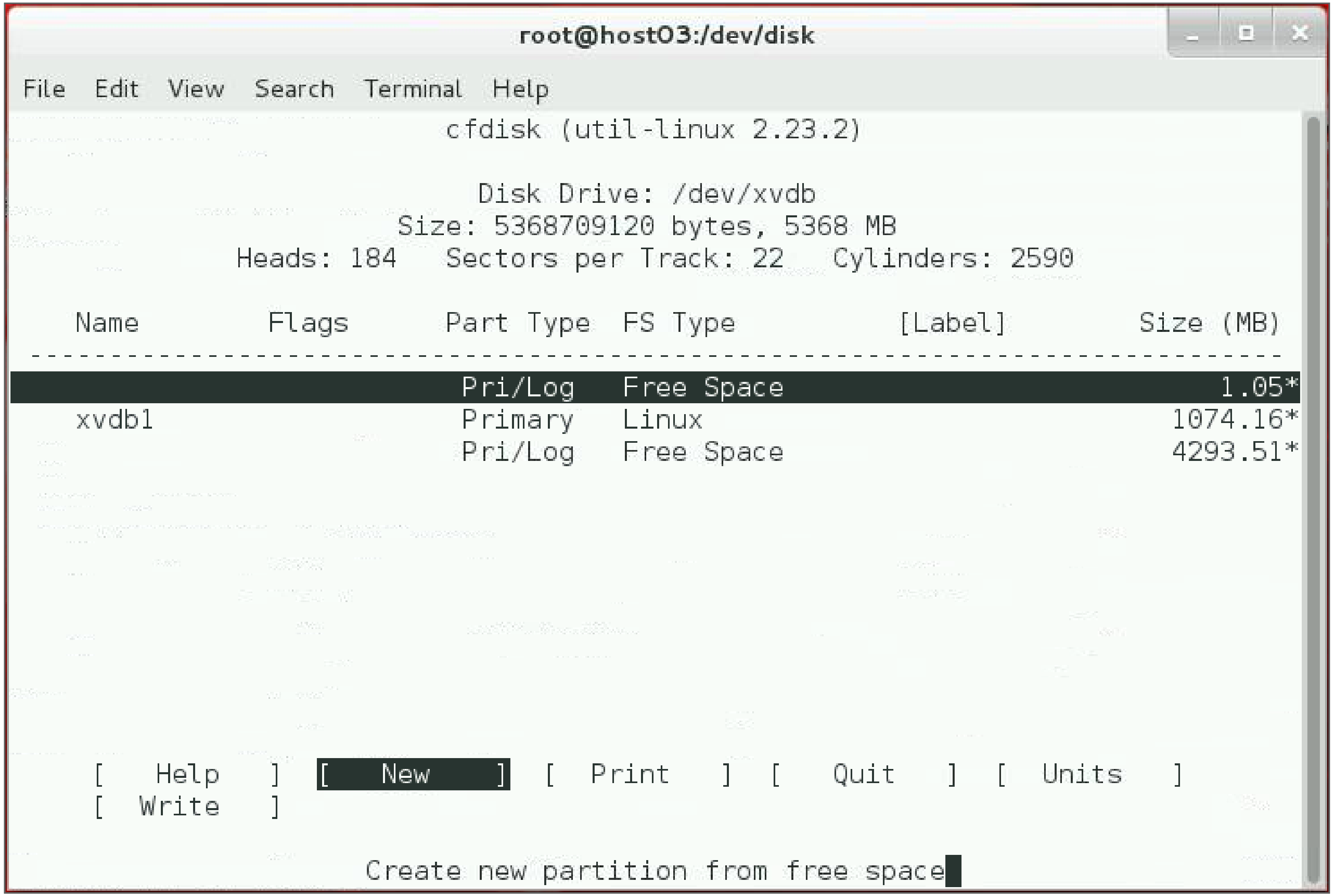
- The `fdisk` utility provides an interactive interface.

```
# fdisk <device_name>
Command (m for help):
```

- Basic `fdisk` commands include:
  - `d`: Delete a partition.
  - `l`: List the known partition types.
  - `m`: Print the available commands.
  - `n`: Add a new partition.
  - `p`: Print the partition table.
  - `q`: Quit without saving changes.
  - `w`: Write the table to disk and exit `fdisk`.
- To have the kernel re-read the partition table:

```
# partprobe <device_name>
```

The `cfdisk` Utility



The `parted` Utility

- The `parted` utility provides a command-line interface:

```
# parted [option] <device_name> [command [argument]]
```

- The `parted` utility also has an interactive mode:

```
# parted <device_name>
(parted)
```

- Interactive mode displays a `(parted)` prompt.
  - Enter `help` to view a list of available commands.
  - Enter `help` *`command`* to view detailed help on a specific command.

File System Types

- **ext2:** High performance for fixed disk and removable media
- **ext3:** Journaling version of ext2
- **ext4:** Supports larger files and file system sizes
- **vfat:** MS-DOS file system useful when sharing files between Windows and Linux
- **XFS:** High-performance journaling file system
- **Btrfs:** Addresses scalability requirements of large storage systems

Making File Systems

- Use the `mkfs` command to build a Linux file system:

```
# mkfs [options] <device>
```

- The `mkfs` command is a wrapper for other utilities:
  - `mkfs.ext2`
  - `mkfs.ext3`
  - `mkfs.ext4`
- Default parameters are specified in `/etc/mke2fs.conf`.
- Use the `blkid` command to display block device attributes.
- Use the `e2label` command to display and modify the file system label.

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Mounting File Systems

- Use the `mount` command to attach a device to the directory hierarchy:

```
# mount [option] <device> <mount_point>
```

- Use the device name, the UUID, or the label:

```
# mount /dev/xvdd1 /test
# mount UUID="uuid_number" /test
# mount LABEL="label_name" /test
```

- Use the `–o` flag to specify mount options:

```
# mount –o nouser,ro /dev/xvdd1 /test
```

- Use the `umount` command to unmount file systems:

```
# umount /dev/xvdd1
```

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Block Volumes in Oracle Cloud Infrastructure

- Dynamically provisioned
- iSCSI or paravirtualized attachment types
  - iSCSI - Bare metal or VM
  - Paravirtualized - VM
- Create, attach, connect, and move volumes
- Created with the Create Block Volume function within the Oracle Cloud Infrastructure console
- Attached from the instance details page
- If iSCSI attachment type used, volume is connected to an instance with `iscsiadm` commands
- Volume size affects block volume performance

The `/etc/fstab` File

- The `/etc/fstab` file is the file system mount table. It:
  - Contains all the information needed by the `mount` command
  - Is read at boot time
- When creating a new file system, add a new entry to the file system mount table in the following format:

```
/dev/xvda3   /boot   ext4   defaults   0   0
```

  - The first column is the device to mount.
  - The second column is the mount point.
  - The third column is the file system type.
  - The fourth column specifies mount options.
  - The fifth column is used by the `dump` command.
  - The last column is used by the `fsck` command.

Maintaining File Systems

- The `fsck` command checks and repairs Linux file systems.
  - `fsck` runs at boot time based on configurable parameters.
  - Do not run `fsck` on mounted file systems.
- Use the `tune2fs` command to:
  - Configure the frequency of file system checks
  - Convert ext2 file systems to ext3
  - Adjust file system parameters on ext2, ext3, and ext4 file systems
  - Display current file system parameter values
- Use the `dumpe2fs` utility to print file system information.
- The `debugfs` utility is an interactive file system debugger.

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Access Control Lists (ACLs)

- An ACL provides a more fine-grained access control mechanism than traditional Linux access permissions.
- The file system containing the file or directory must be mounted with ACL support:

```
# mount -t ext3 -o acl /dev/xvdd1 /test
```

- Include the `acl` option in the `/etc/fstab` file:
  - `LABEL=/work /work ext3 acl 0 0`
- There are two types of ACL rules:
  - `access ACLs`: Specify access information for a single file or directory
  - `default ACLs`: Pertain to a directory only. The default access information for any file within the directory that does not have an access ACL.

`getfacl` and `setfacl` Utilities

- Use the `getfacl` utility to display a file's ACL.
  - When a file does not have an ACL, `getfacl` displays the same information as `ls –l`, although in a different format.
- Use the `setfacl` utility to add or modify rules.
- The rules are in the following form:
  - `u:name:permissions`
  - `g:name:permissions`
  - `m:permissions`
  - `o:permissions`
- To add an ACL rule that gives the `oracle` user read and write permission to the `test` file:

```
# setfacl -m u:oracle:rwx test
```

Swap Space

- Swap space is used when there is insufficient RAM.
- Swap space is a partition, a file, or both.
- Use `fdisk`, `cfdisk`, or `parted` to create a swap partition.
- Use `dd` to create a swap file:

```
# dd if=/dev/zero of=/swapfile bs=1024 count=1000000
```

- Use `mkswap` to initialize a swap partition or file:

```
# mkswap {device|file}
```

- Use `swapon` and `swapoff` to enable and disable devices and files for swapping, respectively:

```
# swapon {device|file}
# swapoff {device|file}
```

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Quiz

Which of the following is the file system mount table for Oracle Linux?

a. `/etc/vfstab`

b. `/etc/filesystem`

c. `/etc/fstab`

d. `/boot/filesystem`

Quiz

Which of the following statements are true?

a. Use the `umount` command to unmount file systems.

b. Use the `swapoff` command to disable swap space.

c. File systems must be unmounted and swap partitions must be disabled before repartitioning a disk drive.

d. Do not run `fsck` on mounted file systems.

Summary

In this lesson, you should have learned how to:

- Describe disk partitioning
- Use disk partitioning utilities
- List supported file system types
- Explain file system creation, mounting, and maintenance
- Describe block volumes in Oracle Cloud Infrastructure
- Describe access control lists (ACLs)
- Describe and maintain swap space

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Practice 12: Overview

This practice covers the following topics:

- Listing the current disk partitions
- Partitioning a storage device
- Creating `ext3` and `ext4` file systems
- Implementing access control lists
- Increasing swap space
- Removing partitions and additional swap space